Screening Protocols

Screening will take place over three stages: Title, Abstract, and Full-Text. We are going to perform the Title and Abstract screening concurrently (i.e., if an article passes the Title screen, screen its abstract before proceeding to the next article) and begin the Full-text screening after both title and abstract screening were finished. For what is expected at each stage, see the specific sections in this document.

Before any article was screened, the lead author explained the screening process in person and answered any questions. Together, they screened several articles to ensure there was consistency across screening. At each stage of screening, articles were ultimately screened by a single author (except in the case of `INSUFFICIENT DETAIL` where a second author was involved to settle disagreements).

1.1 Title Screening

Title Screening happens in Covidence. These instructions detail what to do after you have logged into our project.

For the top article in Covidence:

1. Read the title of the article.
2. In order, check if the title meets any of our exclusion criteria (see Sec 1.4 Exclusion Criteria). E.g., ask yourself "is this article `IRRELEVANT`?" If it seems relevant, ask yourself "is this article a `NON-ARTICLE`?". Continue through the exclusion criteria list asking sequentially if the article's title meets an exclusion criterion.
    a. **NOTE**: at the Title Screen stage, you do not need to check if the Title meets `INSUFFICIENT DETAIL`. This criterion is reserved only for the full text screening.
3. If an article meets any of our exclusion criteria, leave a comment containing "TITLE: [exclusion reason]" in Covidence and mark it as irrelevant in the software.
    a. **NOTE**: the reported exclusion criteria should be the first exclusion criteria the title met as the exclusion criteria should be applied hierarchically.
    b. **Some common reasons an article may be excluded at the Abstract Screen stage:** The title...
        i. Mentions non-human entities for recognition (e.g., sheep or cattle) → `IRRELEVANT`
        ii. Describes a smart-phone only HAR system → `NOT WAO`
        iii. Describes activity recognition not in the scope of behavioral health (e.g., nurse care activities, abnormal activity, indoor localization) → `OOS CT`
        iv. Mentions using only a modality other than acceleration → `NOT WAO`
        v. Uses the word "Survey" or "Review" → `SURVEY/REVIEW`
4. If an article's title does not meet any of our exclusion criteria, send it to the abstract screen.
5. If you are unsure if the title meets the criteria, or believe it is "too vague" to know what the article is about, send it to the abstract screen.

1.2 Abstract Screening

Abstract Screening happens in Covidence immediately after Title screening. If you have sent an article to Abstract Screening from Title Screening, you should perform the Abstract Screening. Abstract Screening is almost identical to performing a Title Screen.

For the top article in Covidence:

1. Read the entire abstract of the article.
2. In order, check if the abstract meets any of our exclusion criteria (see Sec 1.4 Exclusion Criteria). E.g., ask yourself "is this article `IRRELEVANT`?" If it seems relevant, ask yourself "is this article a

`NON-ARTICLE`?". Continue through the exclusion criteria list asking sequentially if the article's abstract meets an exclusion criterion.

   a. **NOTE**: at the Abstract Screen stage, you do not need to check if the Abstract meets `INSUFFICIENT DETAIL`. This criterion is reserved only for the full text screening.

3. If an article meets any of our exclusion criteria, leave a comment containing "ABSTRACT: [exclusion reason]" in Covidence and mark it as irrelevant in the software.
   a. **NOTE**: the reported exclusion criteria should be the first exclusion criteria the abstract met as the exclusion criteria should be applied hierarchically.
   b. **Some common reasons an article may be excluded at the Abstract Screen stage:** The abstract...
      i. Mentions sensor fusion → `NOT WAO`
      ii. Describes explicit sensor location without mentioning the wrist → `NOT WAO`
      iii. Describes using a modality from a non-wearable sensor (e.g., depth video, LIDAR) → `NOT WEARABLE`
      iv. Describes using a thresholding method → `IRRELEVANT`
4. If an article's abstract does not meet any of our exclusion criteria, send it to the Full-Text Screen.
5. If you are unsure if the abstract meets the criteria, or believe it is "too vague" to know what the article is about, send it to the Full-Text Screen.

1.3 Full-Text Screening

Full-Text Screening happens in Covidence. These instructions detail what to do after you have logged into our project.

For the top article in Covidence:

1. Following the article link to open the article from the publisher.
2. If the article is not available via our institution, has been retracted, or is not in English → `NOT AVAILABLE`
3. Go to the "Results" section. If there are no metrics reporting the results of training and validating a HAR model → `IRRELEVANT`
4. If the artifact is a poster or extended abstract → `NOT ARTICLE`
5. Skim the "Methods" section. If the article is a review → `SURVEY/REVIEW`
6. Once you are sure the article is available, relevant, and an actual article, go through each of the exclusion criteria (in order) and ensure the article does not meet any of these criteria.
   a. **NOTE**: Articles normally have sections titled "Data" or "Dataset" that include information on the data they used and should explicitly mention if they used accelerometer data from the wrist sensor. This information can sometimes also be found in a figure detailing the data used as model input.
   b. **NOTE**: Articles oftentimes have tables detailing their activity list. Other times, these activities can be found in the body of the text, or in the results section.
   c. **NOTE**: the reported exclusion criteria should be the first exclusion criteria the abstract met as the exclusion criteria should be applied hierarchically.
7. If an article meets any of our exclusion criteria, leave a comment containing "ABSTRACT: [exclusion reason]" in Covidence and mark it as irrelevant in the software.
   a. **NOTE**: the reported exclusion criteria should be the first exclusion criteria the abstract met as the exclusion criteria should be applied hierarchically.
   b. **Some common reasons an article may be excluded at the Full-Text Screen stage:** The Full-Text...
      i. Describes the use of data only from a non-wrist location → `NOT WAO`
      ii. Describes the use of data only from a non-accelerometer source → `NOT WAO`
      iii. Describes the use of data only from a non-wearable sensor (e.g., depth video, LIDAR) → `NOT WEARABLE`

        iv. Describes using a thresholding method → `IRRELEVANT`
    c. When to use `INSUFFICIENT DETAIL`: The authors of the article describe a known dataset that contains wrist-accelerometer data but do not *explicitly state* that this is the only data used.
        i. Example: The WISDM 2019 dataset contains wrist and phone data. If the authors say they used the WISDM 2019 dataset but do not specific from which location, we *cannot assume* they used data only from the wrist sensor.
        ii. **NOTE**: Articles marked as `INSUFFICIENT DETAIL` will be reviewed by a second author and you two will discuss the proper coding of these articles together.
8. Once you are sure the article does not meet any of the exclusion criteria, ensure it meets each of our inclusion criteria (it should).
    a. If the article does not meet inclusion criterion (1) → `NOT AVAILABLE`
    b. If the article does not meet inclusion criterion (2) → `OOS CT`
    c. If the article does not meet inclusion criterion (3) → `IRRELEVANT`, `NOT WEARABLE`, or `NOT WAO`
9. If an article's Full-Text meets all of our inclusion criteria and none of our exclusion criteria, mark it to be included in the review.

1.4 Exclusion Criteria

The following are our exclusion criteria which should be applied *in order* at each screening stage:

- `IRRELEVANT`: The article is not about human activity recognition (i.e., using a machine learning or deep learning model to classify a set of activities). Articles that do not use a HAR model (e.g., instead use cutpoint/thresholding methods) or do not describe the evaluation of a HAR model were also classified as `IRRELEVANT`.
- `NON-ARTICLE`: The article is published as a poster, workshop paper, extended abstract, or tutorial demo. These types of articles are less likely to have gone through rigorous peer review before publication and are more likely to be a "work-in-progress" without rigorous evaluation acceptable to health researchers. Such articles are thus deemed outside the scope of our review, which is intended to identify HAR models that health surveillance researchers might trust and readily use.
- `SURVEY/REVIEW`: The article is a survey or review of existing work in HAR and does not introduce or describe results of a HAR algorithm.
- `OOS CT` (out-of-scope classification task): The article describes an algorithm capable only of classifying:
    - (a) Sub-activities within a single sport (e.g., different swimming strokes or basketball moves).
    - (b) Sub-activities within a single daily activity16 (e.g., an article describing an algorithm trained to detect "buckling seat belt" and "turning steering wheel" and "driving" is excluded but an algorithm capable of identifying the above activities as well as additional activities like "walking" and "sitting" is included. Additionally, articles describing algorithms capable of classifying only different gaits or walking speeds are excluded).
    - (c) Upper body movements (e.g., arm/hand gestures).
    - (d) Lower body movements (e.g., leg raises or side steps).
- `NON-WEARABLE`: The publication described a HAR algorithm that used data from non-wearable sensors, such as cameras.
- `NOT WAO`: The publication describes and uses data comprised of non-acceleration data from a wrist-worn sensor. The data used may be multimodal, from a non-wrist sensor, from multiple sensors, or any combination.
- `INSUFFICIENT DETAIL`: The article did not include the necessary details (e.g., activity types or sensor locations) to be deemed as included. Each of these articles was screened by at least two authors who both agreed there was insufficient detail in the writing of the article.
- `NOT ARTICLE`: The article was either not available, retracted, or not written in English

1.5 Inclusion Criteria

The following are our inclusion criteria which should be applied *in order* as a final test in the Full-Text Screen:

(1) Be written in English.

(2) Summarize at least one algorithm to recognize an individual doing at least two distinct 1) physical activities, including at least one clear ambulation activity (e.g., walking) or a clear sedentary behavior (e.g., lying down or sitting) or 2) estimated physical activity intensity levels (i.e., sedentary, light, moderate, vigorous).

(3) Evaluate at least one summarized algorithm on a single wrist-accelerometer-only data stream.